

Solution to Hidoku Queens

Serena Chen

Preface

This solution mainly contains strategies on how to solve the puzzle. There are many ways to solve the puzzle aside from the ones mentioned, but there will only be one unique solution to each part of the puzzle.

Solution

Secret Word: ANALYSIS

4	2	1	10	38	39	40	41
3	5	9	11	37	19	20	42
6	8	13	12	36	18	21	43
7	14	15	35	17	22	46	44
32	33	34	16	23	49	47	45
31	26	25	24	51	50	48	64
27	30	54	52	56	59	60	63
28	29	53	55	58	57	61	62

1 Solving the Hidoku

Starting with the number 1, we can follow the forced path of consecutive numbers by only connecting the adjacent cells (including diagonals). We can notice that there are certain sequences that constrain the solution. For example, the numbers 1-10 must be in the top left region. The numbers from 30-40 start from the bottom left region and slowly makes its way towards the top right region, and the puzzle must end in the bottom right region. We can start by filling out given adjacent numbers and eliminating possible squares for numbers. Some of these examples include 1-5, 28-32, and 46-49. By looking at forced placements and sequences, we can eliminate impossible choices and solve the puzzle. It is easier to use logical reasoning to solve the puzzle instead of trial and error!

2 Queens Crown Placement

The following solution is based on the assumption that part 1 has already been solved. Box + number refers to the box that has that number in it. For example, box 23 refers to the box that has the number 23 in it.

- **Step 1:** We can notice that there must not be a crown in where the numbers 9, 20 and 26 are, as a crown in those places would eliminate all the boxes in the orange, yellow and red regions respectively.
- **Step 2:** Since the crowns placed in the orange and yellow regions must occupy the top two rows, we can cross out any boxes that are not orange or yellow in those rows. This leaves one box in the blue region, box 12. Place a crown there.
- **Step 3:** We can cross out all the boxes that are in the same row and column as the crown placed in box 12, as well as the ones that touch it diagonally. We can also notice that the crown in the left-most column must be in the red region, so we can cross out any boxes in that column that are not red. This leaves box 14 in the green region where there must be a crown.
- **Step 4:** After crossing out the boxes in the same row and column as well as touching diagonally to the crown placed in box 14, this leaves box 1 in the orange region. Place a crown there and cross out all the necessary boxes.
- **Step 5:** After crossing out all the necessary boxes, we are left with box 19 in the yellow region and box 51 in the turquoise region.
- **Step 6:** Repeat the process of crossing out the necessary boxes to be left with box 27 in the red region, box 61 in the pink region, and box 45 in the purple region.

3 Decoding the Secret Word

After part 2, the following numbers should be circled: 27, 14, 1, 12, 51, 19, 61, 45. By taking modulo 26 of the numbers and converting them to their respective letters, we get

$$27 = A$$

$$14 = N$$

$$1 = A$$

$$12 = L$$

$$51 = Y$$

$$19 = S$$

$$61 = I$$

$$45 = S$$

Going from left to right, we get the secret word **ANALYSIS**.